

COL724/7524 Advanced Computer Networks

Assignment 1 – Part D Failure and Recovery Study

1 Objective

Part D studies the behavior of the network when the direct AS1–AS4 link carrying traffic is temporarily unavailable. The experiment is based on the Part B topology, which contains the three-hop path

$$\text{AS1} \rightarrow \text{AS2} \rightarrow \text{AS3} \rightarrow \text{AS4}$$

and the direct path

$$\text{AS1} \rightarrow \text{AS4}.$$

The direct AS1–AS4 link is selected during normal operation because it provides a shorter and lower-cost path. The link is failed at $t = 6$ s and restored at $t = 9$ s. The objective is to observe whether traffic continues through the alternate path and to measure packet delivery, delay, and jitter before, during, and after the failure.

2 Experimental Setup

The Part D simulation uses the same link configuration as Part B. The direct AS1–AS4 link has a data rate of 11 Mbps and a propagation delay of 1 ms, while the three-hop path consists of links with delays of 3 ms, 4 ms, and 3 ms, respectively.

UDP traffic is generated from AS1 toward AS4. The FlowMonitor module records the transmitted and received packets, packet loss, delay, and jitter. At $t = 6$ s, the AS1–AS4 interface is brought down and the routing tables are recomputed. At $t = 9$ s, the interface is brought up again and the routing tables are recomputed once more.

The three observation intervals are therefore:

- Before failure: $0 \leq t < 6$ s
- During failure: $6 \leq t < 9$ s
- After recovery: $9 \leq t \leq 14$ s

3 Failure and Recovery Timeline

The experiment follows the timeline below.

Table 1: Part D failure and recovery timeline

Time	Event	Expected path
Before 6 s	Normal operation	AS1–AS4
6 s	AS1–AS4 failure	AS1–AS2–AS3–AS4
6–9 s	Failure interval	AS1–AS2–AS3–AS4
9 s	AS1–AS4 recovery	AS1–AS4
After 9 s	Normal operation restored	AS1–AS4

The direct AS1–AS4 link was chosen for failure because it was the active path immediately before the failure. Once this link becomes unavailable, the three-hop path through AS2 and AS3 provides the available alternate route.

4 FlowMonitor Results

The FlowMonitor output was examined separately for the three failure intervals. The direct-path flow contains twelve successfully received packets, while the alternate-path flow during the failure contains four successfully received packets. Both flows report zero lost packets. The alternate-path flow also has non-zero forwarding events, consistent with packets being forwarded through AS2 and AS3.

Table 2: FlowMonitor results for the three intervals

Interval	Path	Tx	Rx	Lost	Delivery
Before failure	AS1–AS4	6	6	0	100%
During failure	AS1–AS2–AS3–AS4	4	4	0	100%
After recovery	AS1–AS4	6	6	0	100%

Thus, the failure does not cause a complete outage because an alternate path is available. All packets recorded by FlowMonitor are successfully delivered.

The average delay for each path is calculated from the FlowMonitor attributes using

$$D_{\text{avg}} = \frac{\text{delaySum}}{\text{rxPackets}}.$$

For the direct-path flow,

$$D_{\text{direct}} = \frac{1.67302 \times 10^7 \text{ ns}}{12} \approx 1.3942 \text{ ms.}$$

For the alternate-path flow,

$$D_{\text{alternate}} = \frac{4.72267 \times 10^7 \text{ ns}}{4} \approx 11.8067 \text{ ms.}$$

The FlowMonitor jitter values are zero for the measured flows.

Table 3: Delay and jitter measurements

Interval	Path	Average Delay (ms)	Jitter (ms)
Before failure	AS1–AS4	1.3942	0
During failure	AS1–AS2–AS3–AS4	11.8067	0
After recovery	AS1–AS4	1.3942	0

The alternate path therefore increases the measured FlowMonitor delay by

$$11.8067 - 1.3942 = 10.4125 \text{ ms.}$$

The higher delay is expected because packets traverse three links instead of the single direct AS1–AS4 link.

5 Delay Variation with Time

Figure 1 shows the delay measured from the packet-level CSV over the simulation time. The vertical markers indicate the failure at $t = 6$ s and recovery at $t = 9$ s.

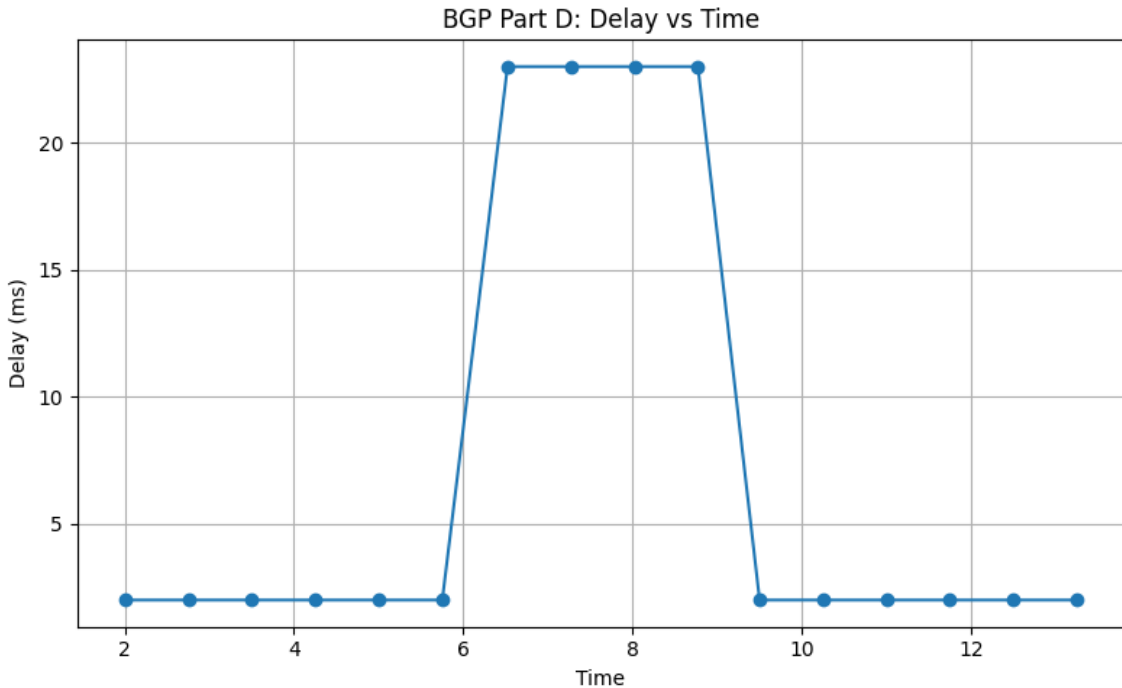


Figure 1: Measured packet delay during AS1–AS4 failure and recovery.

Before $t = 6$ s, the measured delay remains at the lower direct-path level. Immediately after the AS1–AS4 failure, the delay increases because traffic is carried over the longer AS1–AS2–AS3–AS4 path. The higher-delay interval continues until the direct link is restored at $t = 9$ s. After recovery, the delay returns to the original lower level.

The packet-level CSV used for this figure records the time between UDP client transmission and receipt of the Echo response. Therefore, its values are an RTT-style measurement, whereas the FlowMonitor delay values in Table 3 are the flow-level end-to-end

delay reported by FlowMonitor. The two measurements show the same qualitative behavior: the alternate path produces a much larger delay and the delay returns to its previous level after recovery.

6 Route Change Analysis

The routing behavior changes at both link-state events.

At $t = 6$ s, the direct AS1–AS4 interface is disabled. The direct route is therefore no longer available and traffic is forwarded through

$$\boxed{\text{AS1} \rightarrow \text{AS2} \rightarrow \text{AS3} \rightarrow \text{AS4}}$$

during the failure interval.

The FlowMonitor XML supports this route change through the forwarding statistics. The alternate-path flow has four transmitted and four received packets and reports eight forwarding events. Since each packet is forwarded through two intermediate routers, the forwarding count is consistent with the AS1–AS2–AS3–AS4 path.

At $t = 9$ s, the AS1–AS4 interface is restored and the routing tables are recomputed. Traffic subsequently returns to the direct AS1–AS4 path.

Thus, the complete route transition is

$$\text{AS1–AS4} \xrightarrow{t=6s} \text{AS1–AS2–AS3–AS4} \xrightarrow{t=9s} \text{AS1–AS4}.$$

7 Conclusion

Part D demonstrates link failure and recovery in the four-router inter-AS topology. The direct AS1–AS4 link was failed at $t = 6$ s and restored at $t = 9$ s. During the failure, traffic successfully used the alternate AS1–AS2–AS3–AS4 path.

FlowMonitor recorded 100% packet delivery in all three intervals. The average end-to-end delay increased from approximately 1.3942 ms on the direct path to 11.8067 ms on the alternate path, and returned to 1.3942 ms after recovery. The measured jitter was zero for the recorded flows.

Overall, the experiment shows that the presence of an alternate path allows the network to remain connected during a link failure, at the cost of increased end-to-end delay.